Rutul Shah

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SUMMARY

Full Stack Developer specializing in the MERN stack, with a robust background in both frontend and backend development. Expertise in real- time data streaming, infrastructure automation, and monitoring/alerting solutions. Known for strong collaboration and communication skills, dedicated to advancing best practices and fostering a culture of continuous improvement. Proven track record of designing and delivering high- quality, scalable software solutions in Agile environments.

EXPERIENCE

Lead Full Stack Developer (Associate Director), Multi Asset Trade Analytics

Royal Bank of Canada **January 2026 - Present**

- **Leading development** of a multi-asset trade analytics platform enabling RBC traders and external clients to analyze pre-trade insights, Index Analytics, FX price movement (FXPM), and News Sentiment data
- Part of RBC's **AIDLC initiative**, evolving the traditional SDLC Methodology to embed stakeholders and AI at every step of the delivery process, accelerating how solutions are delivered
- Developing **responsive** and **high-performance** user interfaces using React and Next.js to visualize complex financial datasets and trading insights.
- Building **scalable backend services** using Nest.js and Python to support market analytics and trade intelligence workflows.
- **Architecting real-time data communication** using multiple protocols (REST, WebSockets, SSR), selecting optimal patterns based on latency, scalability, and business requirements.
- **Collaborating with traders, product managers, and engineering teams** to translate complex trading requirements into technical solutions.
- Participating in **architecture discussions, code reviews, and** enforcing **engineering best practices** as part of a lead development role while **mentoring** and promoting **clean architecture** and **scalable design patterns**.
- **Supporting production systems** through monitoring, troubleshooting, and continuous performance improvements.

Full Stack Engineer, Small Business Center

Paymentus **November 2021 – January 2026**

- **Engineered and deployed responsive web applications** utilizing React, TypeScript, and Redux, delivering optimal user experiences with the Ant Design (Antd) UI framework.
- **Developed robust backend services** using Node.js and Express.js, integrating with MongoDB and Oracle databases to support dynamic data handling and storage.
- **Designed, developed, and integrated** RESTful APIs to support seamless communication between microservices, enabling scalable and efficient data exchange across distributed systems.
- **Architected real-time data streaming pipelines** with Apache Kafka, enhancing communication and synchronization across microservices for improved system efficiency.
- **Implemented comprehensive monitoring and alerting** solutions with Prometheus and Grafana, facilitating proactive issue detection and resolution to maintain system reliability.
- **Managed infrastructure and deployment processes** with Terraform, Helm Charts, and ArgoCD, ensuring scalability, high availability, and streamlined application deployment.
- **Streamlined continuous integration and delivery** (CI/CD) pipelines using Jenkins and DroneCI, accelerating development cycles and ensuring reliable, frequent releases.
- **Collaborated with cross-functional teams** in an agile setting, actively participating in sprint planning, daily stand-ups, and retrospectives to drive project success.
- **Conducted thorough code reviews** and provided actionable feedback to peers, promoting adherence to best practices and maintaining high code-quality standards.
- Collaborated on the **design, implementation,** and **optimization** of diverse payment solutions, including Digital Wallets, Scheduled Payments, ACH Payments, Future-Dated Payments, and Card Payments, ensuring seamless user experiences and secure transaction processing.

Software Engineer, Financial Crime

Royal Bank of Canada **July 2020 - November 2021**

- **Developed and maintained a Time Series Historical Database** to generate trade alerts based on global trading activities conducted by RBC traders, enhancing real-time monitoring and response.
- **Engineered a Data Loader tool** to automate the ingestion of files from diverse sources into the database, streamlining data integration and improving operational efficiency.
- **Integrated new RESTful APIs** to enrich database content with Market Research Data (MRD), providing valuable insights and enhancing data accuracy.
- **Led migration efforts** Run the Bank (RTB) team, facilitating smooth transitions and minimizing disruption during system upgrades.
- **Created an interactive dashboard** for support teams to monitor trading alerts and identify suspicious activities, enabling prompt investigation and resolution of potential issues.
- **Assisted the QA team in the testing process** by providing comprehensive details and documentation for newly developed features, ensuring thorough validation and quality assurance.

Production Support Specialist, Mainframe Systems

Royal Bank of Canada **January 2018 - July 2020**

- Automated the monthly statistics gathering process for the team, creating VBA/Python scripts to minimize manual work and presenting data through Excel charts.
- Identified and addressed process errors in collaboration with management, implementing effective solutions to prevent recurrence.
- Mentored and onboarded new hires, guiding them in best practices and operational procedures.
- Monitored and ensured compliance with Service Level Agreements (SLAs) by resolving production batch failures promptly and documenting potential impacts to the bank.
- Managed migration, administration, and development of Elasticsearch applications.
- Optimized batch processing schedules across multiple mainframe systems, reducing average job completion time by 25% and enhancing system efficiency and resource utilization.

Software Engineer, Co-op

Royal Bank of Canada **September 2017 - December 2017**

- Contributed to gathering data from internal servers creating various APIs using Vanilla JavaScript, AngularJS, Express.js, ELK Stack (Elastic Search, Logstash and Kibana) and JSON objects.
- Developed path-finding algorithm so that application could prioritize and perform data authenticity based on server.
- Automated the manual process which saved 3 weeks of manual work every quarter.
- Improved user experience in monthly newsletters to make it responsive and user friendly.

Software Engineer, Co-op

TD Bank **September 2016 - December 2016**

- Built bar and pie charts which would increase its value showing the amount saved by team every second using JavaScript, Perl and jQuery.
- Attended team meetings and scrum calls on a regular basis following agile methodology.

EDUCATION

Diploma in Computer Programming

Georgian College • Barrie, Ontario **January 2016- December 2017**

Bachelor of Engineering in Computer Science

Gujarat Technological University • India **May 2011 – June 2015**

SKILLS

HTML | CSS | JavaScript | TypeScript | React | Redux | Next.js | Nest.js | Node.js (Express.js) | MongoDB | Kafka | Docker | Kubernetes | Terraform | Helm | ArgoCD | Amazon Web Services (AWS) | Grafana | Prometheus | InfluxDB